

T.E./Insem. - 152
T.E. (IT)
THEORY OF COMPUTATION
(2012 Pattern) (Semester - I)

Time : 1 Hour]

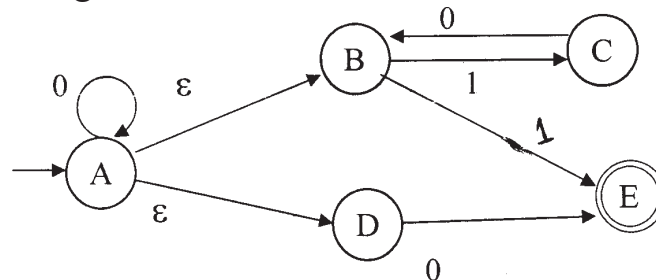
[Max. Marks : 30

Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

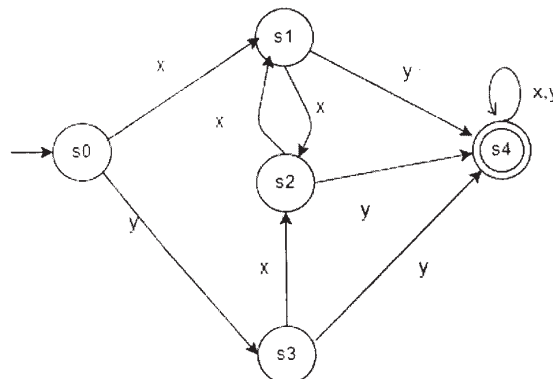
Q1) a) Construct a deterministic finite automata (DFA) that recognizes the language $L = \{x \in (0,1) \mid (x \text{ contains atleast two consecutive 0's) and } (x \text{ does not contain two consecutive 1's})\}$ **[4]**

b) Convert the given NFA- ϵ to an NFA. **[6]**



OR

Q2) a) Minimize the following automata. **[6]**



b) Convert following Moore machine into Mealy Machine. **[4]**

P.T.O.

Present State	Next State		Output
	a = 0	a = 1	
-> q0	q3	q1	1
q1	q2	q3	0
q2	q2	q1	0
q3	q1	q0	1

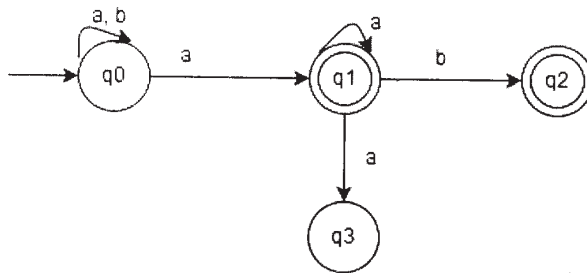
Q3) a) Find the regular expression corresponding to each of the following subset of $\{0, 1\}$ [4]

- Language of all Strings not containing the substring 000.
- Language of all Strings containing an even no of 0's.

b) Show that $L = \{a^n b^{2n} \mid n > 0\}$ is not regular. [6]

OR

Q4) a) Construct a regular expression for given finite automata. [6]



b) Convert Right Linear Grammar to equivalent Left Linear grammar [4]

$S \rightarrow bB$

$B \rightarrow bC$

$B \rightarrow aB$

$C \rightarrow a$

$B \rightarrow b$

Q5) a) Convert the following grammar in Chomsky NF [8]

$S \rightarrow 01XY$

$X \rightarrow 1XY \mid \epsilon$

$Y \rightarrow YX0 \mid X \mid \epsilon$

b) Define Context free Grammar with example. [2]

OR

- Q6)** a) Write a context free grammar for following Language [6]
- i) $L = \{a^m b^n c^p \mid m+n=p\}$
 - ii) $(a+b)b(a+b)^*b(a+b)$
- b) Write short note on Chomsky Hierarchy. [4]

